CHRISTOPHER JAMES MATTHEWS

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Bristol, United Kingdom

PROFESSIONAL SUMMARY

I'm a Senior Machine Learning Engineer with six years of hands-on experience building and deploying production ML systems across fintech and healthcare sectors. I've worked through the full lifecycle of model development from initial research through to monitoring in production, and I'm particularly interested in scaling ML infrastructure and making models interpretable for non-technical stakeholders. I thrive in environments where I can mentor junior engineers while pushing the boundaries of what's technically possible.

WORK EXPERIENCE

Machine Learning Engineer - Staff Level

Vertex Intelligence Ltd

vertexintel.io

London, United Kingdom

January 2023 - Present

Architected and deployed a real-time fraud detection system processing 50 million transactions daily using gradient boosting and deep learning ensembles. Managed a cross-functional team of 22 engineers including backend developers, data engineers, and ML specialists across three continents. Reduced model inference latency by 65% through quantization and implemented automated retraining pipelines using GPT-4 API for prompt-based feature engineering. Established company-wide ML best practices and led the migration to a microservices architecture that enabled 15 simultaneous model experiments in production.

Principal Research Engineer

NeuralPath Systems

neuralpathsys.co

Bristol, United Kingdom

September 2021 - December 2022

Led a team of 18 data scientists in developing computer vision models for medical imaging analysis, achieving 94.2% accuracy on pneumonia detection across multiple radiograph types. Implemented reinforcement learning solutions for dynamic resource allocation in hospital networks, resulting in 23% improvement in patient throughput. Pioneered the adoption of LangChain for generating clinical decision support explanations and developed a multi-modal transformer architecture combining text and image data. Managed annual budget allocation of £2.3 million and reported directly to the VP of Research on strategic ML initiatives.

Junior Machine Learning Engineer

DataFlow Analytics

dataflowanalytics.net

Bristol, United Kingdom

June 2019 - August 2021

Developed and maintained time series forecasting models for energy consumption predictions using LSTM networks and AutoML frameworks. Implemented GitHub Actions workflows for automated model testing and CI/CD pipelines, deploying weekly updates to production systems. Built data ingestion pipelines processing 500GB daily

from IoT sensors and developed custom Python packages for feature engineering. Collaborated with business analysts to translate requirements into model specifications and provided technical support for deployed models serving 40+ enterprise clients.

EDUCATION

MSc Computer Science with Specialisation in Machine Learning

University of Bristol

Bristol, United Kingdom

2017 - 2019

BSc Mathematics with Computer Science

University of Warwick

Coventry, United Kingdom

2013 - 2016

TECHNICAL SKILLS

Machine Learning: TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM, Deep Learning, Reinforcement Learning, Computer Vision, Natural Language Processing, Time Series Analysis

Data Processing: Python, Pandas, NumPy, PySpark, SQL, Apache Spark, ETL pipelines

ML Operations: Model deployment, A/B testing, Monitoring and alerting, Model versioning, MLflow, Kubernetes, Docker

Cloud Platforms: AWS (SageMaker, EC2, S3), Google Cloud Platform, Azure ML

Specialisations: Fraud detection, Medical imaging, Forecasting, Ensemble methods, Feature engineering, Model interpretability

CERTIFICATIONS

AWS Certified Machine Learning - Specialty

Google Cloud Professional Data Engineer